

Assignment 8

A manager M chooses $a \in \{a_1, a_2\}$ the cost for M of c_i with $c_1 < c_2$

The project yields a profit π_H or π_L

$$\mathbb{P}(\pi = \pi_H | a_1) = \gamma_1 < \gamma_2 = \mathbb{P}(\pi = \pi_H | a_2)$$

The managers utility function is given

$$U(y, c) = \sqrt{y} - c$$

With reservation utility

$$\bar{U} = 0$$

The parameters satisfy:

$$\gamma_2 > \frac{\gamma_1 c_2}{c_1}$$

The owner is risk neutral meaning that we can by affine transformation set $V(\pi) = \pi$

a) We set up and solve the maximisation problem

$$\begin{aligned} \max_{w_H, w_L, a_k} & \gamma_k \cdot (\pi_H - w_h) + (1 - \gamma_k)(\pi_L - w_L) \\ \text{s.t.} & \gamma_k(\sqrt{w_H} - c_k) + (1 - \gamma_k)(\sqrt{w_L} - c_k) \geq 0 \end{aligned}$$

We set a_k konstant and maximise over w_H, w_L

$$\begin{aligned} \max_{w_H, w_L} & \gamma_k \cdot (\pi_H - w_h) + (1 - \gamma_k)(\pi_L - w_L) \\ \text{s.t.} & \gamma_k(\sqrt{w_H} - c_k) + (1 - \gamma_k)(\sqrt{w_L} - c_k) \geq 0 \end{aligned}$$

We notice that V is linear with positive coefficients and thus concave

Also the sum of concave functions is concave thus the constraint is also concave

Thus optimisation becomes nice and easy

We set up the lagrangian

$$\mathcal{L} = \gamma_k \cdot (\pi_H - w_h) + (1 - \gamma_k)(\pi_L - w_L) + \lambda [\gamma_k(\sqrt{w_H}) + (1 - \gamma_k)(\sqrt{w_L}) - c_k]$$

$$\frac{\partial \mathcal{L}}{\partial w_H} = -\gamma_k + \lambda \cdot \frac{\gamma_k}{2 \cdot \sqrt{w_H}} = 0. \quad (1)$$

$$\frac{\partial \mathcal{L}}{\partial w_L} = -(1 - \gamma_k) + \lambda \cdot \frac{1 - \gamma_k}{2 \cdot \sqrt{w_L}} = 0. \quad (2)$$

$$\lambda \cdot \frac{\partial \mathcal{L}}{\partial \lambda} = 0 \Leftrightarrow \lambda = 0 \quad \text{or} \quad \gamma_k(\sqrt{w_H}) + (1 - \gamma_k)(\sqrt{w_L}) = c_k$$

$$\text{AFC } \gamma_k(\sqrt{w_H}) + (1 - \gamma_k)(\sqrt{w_L}) > c_k \quad \text{and maximises}$$

$$\gamma_k \cdot (\pi_H - w_h) + (1 - \gamma_k)(\pi_L - w_L)$$

$$\text{since } \sqrt{x} \text{ is continuous } \exists \epsilon > 0 : \gamma_k(\sqrt{w_H} - \epsilon) + (1 - \gamma_k)(\sqrt{w_L}) > c_k$$

With payoff

$$\gamma_k \cdot (\pi_H - w_h + \epsilon) + (1 - \gamma_k)(\pi_L - w_L) > \gamma_k \cdot (\pi_H - w_h) + (1 - \gamma_k)(\pi_L - w_L)$$

Thus we must have

$$\gamma_k(\sqrt{w_H}) + (1 - \gamma_k)(\sqrt{w_L}) = c_k \quad (3)$$

In optimum

$$\begin{aligned}
(1) &\Leftrightarrow \lambda = 2\sqrt{w_H} \\
(2) &\Leftrightarrow \lambda = 2\sqrt{w_L} \\
&\Rightarrow w_H = w_L = w \\
(3) &\Rightarrow \sqrt{w} = c_K \Rightarrow w = c_K^2
\end{aligned}$$

Now we check when which is best

c_1 :

$$\gamma_1 \cdot (\pi_H - c_1^2) + (1 - \gamma_1)(\pi_L - c_1^2) = \gamma_1 \cdot \pi_H + (1 - \gamma_1)\pi_L - c_1^2$$

c_2 :

$$\gamma_2 \cdot (\pi_H - c_2^2) + (1 - \gamma_2)(\pi_L - c_2^2) = \gamma_2 \cdot \pi_H + (1 - \gamma_2)\pi_L - c_2^2$$

When should we choose a1?

$$\begin{aligned}
&\gamma_1 \cdot \pi_H + (1 - \gamma_1)\pi_L - c_1^2 \geq \gamma_2 \cdot \pi_H + (1 - \gamma_2)\pi_L - c_2^2 \\
&\Leftrightarrow \pi_H(\gamma_1 - \gamma_2) + \pi_L(\gamma_2 - \gamma_1) = (\pi_H - \pi_L)(\gamma_1 - \gamma_2) \geq c_1^2 - c_2^2
\end{aligned}$$

Otherwise we choose a2

b) :

when implimenting a_i the equation must satisfy

$$\begin{aligned}
&\gamma_i(\sqrt{w_H}) + (1 - \gamma_i)(\sqrt{w_L}) - c_i \geq \gamma_{-i}(\sqrt{w_H}) + (1 - \gamma_{-i})(\sqrt{w_L}) - c_{-i} \\
&\Leftrightarrow \sqrt{w_H}(\gamma_i - \gamma_{-i}) - \sqrt{w_L}(\gamma_i - \gamma_{-i}) + c_{-i} - c_i = (\sqrt{w_H} - \sqrt{w_L})(\gamma_i - \gamma_{-i}) + c_{-i} - c_i \geq 0
\end{aligned}$$

Where $-i$ denotes the other option

c) :

We set up the problem (keeping k as general so we dont have to do it twice)

$$\begin{aligned}
&\max_{w_H, w_L} \gamma_i \cdot (\pi_H - w_h) + (1 - \gamma_i)(\pi_L - w_L) \\
&s. t. \quad \gamma_i(\sqrt{w_H}) + (1 - \gamma_i)(\sqrt{w_L}) - c_i \geq 0 \\
&\quad (\sqrt{w_H} - \sqrt{w_L})(\gamma_i - \gamma_{-i}) + c_{-i} - c_i \geq 0
\end{aligned}$$

I will argue that these must both hold with equality and then solve the system of equations

$$\begin{aligned}
&\gamma_i\sqrt{w_H} + (1 - \gamma_i)\sqrt{w_L} - c_i \geq 0. (1) \\
&(\sqrt{w_H} - \sqrt{w_L})(\gamma_i - \gamma_{-i}) + c_{-i} - c_i \geq 0. (2)
\end{aligned}$$

(1)

Assume we have a soultion (w'_h, w'_L)

With (2) holding and (1) being a strict inequality

Since $\sqrt{x}$ is continous

$$\gamma_i\sqrt{w'_H} + (1 - \gamma_i)\sqrt{w'_L} - c_i > 0$$

$$\begin{aligned}
& \left(\sqrt{w'_H} - \sqrt{w'_L} \right) (\gamma_i - \gamma_{-i}) + c_{-i} - c_i \geq 0 \\
& \quad \exists \delta, \epsilon > 0: \\
& \left(\sqrt{w'_H - \delta} - \sqrt{w'_L - \epsilon} \right) = \left(\sqrt{w'_H} - \sqrt{w'_L} \right) \\
& \text{and } \gamma_i \sqrt{w'_H - \delta} + (1 - \gamma_i) \sqrt{w'_L - \epsilon} - c_i > 0
\end{aligned}$$

But then

$$\gamma_i \cdot (\pi_H - w'_h) + (1 - \gamma_i)(\pi_L - w'_L) < \gamma_i \cdot (\pi_H - w'_h + \delta) + (1 - \gamma_i)(\pi_L - w_L + \epsilon)$$

And thus (w'_H, w'_L) cannot be a solution

And we must have

$$\gamma_i \sqrt{w'_H} + (1 - \gamma_i) \sqrt{w'_L} - c_i = 0$$

(2)

Assume (w'_H, w'_L) is a solution and

$$\begin{aligned}
& \gamma_i \sqrt{w'_H} + (1 - \gamma_i) \sqrt{w'_L} - c_i \geq 0 \\
& \left(\sqrt{w'_H} - \sqrt{w'_L} \right) (\gamma_i - \gamma_{-i}) + c_{-i} - c_i > 0
\end{aligned}$$

I could not come up with an argument for this one...

We solve a simple system of equations

$$\begin{aligned}
& \left(\sqrt{w_H} - \sqrt{w_L} \right) (y_i - y_j) + c_j - c_i = 0 \quad (1) \\
& y_i \sqrt{w_H} + (1 - y_i) \sqrt{w_L} - c_i = 0 \quad (2)
\end{aligned}$$

$$\begin{aligned}
(1) & \Leftrightarrow \left(\sqrt{w_H} - \sqrt{w_L} \right) (y_i - y_j) = c_i - c_j \\
& \left(\sqrt{w_H} - \sqrt{w_L} \right) = \frac{c_i - c_j}{(y_i - y_j)} \\
& \sqrt{w_H} = \sqrt{w_L} + \frac{c_i - c_j}{(y_i - y_j)}
\end{aligned}$$

$$\begin{aligned}
(2) & \Rightarrow y_i \sqrt{w_H} + (1 - y_i) \sqrt{w_L} = y_i \left(\sqrt{w_L} + \frac{c_i - c_j}{(y_i - y_j)} \right) + (1 - y_i) \sqrt{w_L} \\
& = \sqrt{w_L} + y_i \cdot \frac{c_i - c_j}{(y_i - y_j)} = c_i
\end{aligned}$$

$$\Leftrightarrow \sqrt{w_L} = c_i - y_i \cdot \frac{c_i - c_j}{(y_i - y_j)} = \frac{c_j \cdot y_i - c_i \cdot y_j}{y_i - y_j}$$

$$w_L = \left(\frac{c_j \cdot y_i - c_i \cdot y_j}{y_i - y_j} \right)^2$$

$$\sqrt{w_H} = \sqrt{w_L} + \frac{c_i - c_j}{(y_i - y_j)} = \frac{c_j \cdot y_i - c_i \cdot y_j}{y_i - y_j} + \frac{c_i - c_j}{(y_i - y_j)} = \frac{c_i \cdot (1 - y_j) - c_j \cdot (1 - y_i)}{y_i - y_j}$$

$$w_H = \left(\frac{c_i \cdot (1 - y_j) - c_j \cdot (1 - y_i)}{y_i - y_j} \right)^2$$

For at implementerer a_2 her er 2 index ikke potens

$$w_H^2 = \left(\frac{c_2 \cdot (1 - y_1) - c_1 \cdot (1 - y_2)}{y_2 - y_1} \right)^2$$

$$w_L^2 = \left(\frac{c_1 \cdot y_2 - c_2 \cdot y_1}{y_2 - y_1} \right)^2$$

d) :

We simply calculate expected profits and evaluate

For 1 we can simply implement the first best since given equal wages the agent will choose strictly lower costs

Reminder that

$$w^1 = w_L^1 = w_H^1 = c_1^2$$

$$\Pi_1 = y_1 \cdot \pi_H + (1 - y_1) \cdot \pi_L - c_1^2$$

$$\Pi_2 = y_2 \cdot \left(\pi_H - \left(\frac{c_2 \cdot (1 - y_1) - c_1 \cdot (1 - y_2)}{y_2 - y_1} \right)^2 \right) + (1 - y_2) \left(\pi_L - \left(\frac{c_1 \cdot y_2 - c_2 \cdot y_1}{y_2 - y_1} \right)^2 \right)$$

We implement first best if and only if

$$\begin{aligned} \Pi_1 &\geq \Pi_2 \Leftrightarrow \\ y_1 \cdot \pi_H + (1 - y_1) \cdot \pi_L - c_1^2 &\geq y_2 \cdot \left(\pi_H - \left(\frac{c_2 \cdot (1 - y_1) - c_1 \cdot (1 - y_2)}{y_2 - y_1} \right)^2 \right) \\ &\quad + (1 - y_2) \left(\pi_L - \left(\frac{c_1 \cdot y_2 - c_2 \cdot y_1}{y_2 - y_1} \right)^2 \right) \\ \Leftrightarrow (\pi_H - \pi_L)(y_1 - y_2) &\geq c_1^2 - y_2 \cdot \left(\frac{c_2 \cdot (1 - y_1) - c_1 \cdot (1 - y_2)}{y_2 - y_1} \right)^2 - (1 - y_2) \left(\frac{c_1 \cdot y_2 - c_2 \cdot y_1}{y_2 - y_1} \right)^2 \\ = c_1^2 - y_2 \cdot \left(\left(\frac{c_2 \cdot (1 - y_1) - c_1 \cdot (1 - y_2)}{y_2 - y_1} \right)^2 - \left(\frac{c_1 \cdot y_2 - c_2 \cdot y_1}{y_2 - y_1} \right)^2 \right) &- \left(\frac{c_1 \cdot y_2 - c_2 \cdot y_1}{y_2 - y_1} \right)^2 \end{aligned}$$

Look at the difference of squares

$$\begin{aligned} &\left(\frac{c_2 \cdot (1 - y_1) - c_1 \cdot (1 - y_2)}{y_2 - y_1} \right)^2 - \left(\frac{c_1 \cdot y_2 - c_2 \cdot y_1}{y_2 - y_1} \right)^2 \\ &= \left(\frac{c_2 \cdot (1 - y_1) - c_1 \cdot (1 - y_2)}{y_2 - y_1} - \frac{c_1 \cdot y_2 - c_2 \cdot y_1}{y_2 - y_1} \right) \\ &\quad \left(\frac{c_2 \cdot (1 - y_1) - c_1 \cdot (1 - y_2)}{y_2 - y_1} + \frac{c_1 \cdot y_2 - c_2 \cdot y_1}{y_2 - y_1} \right) \\ &= \frac{c_2 - c_1}{y_2 - y_1} \cdot \frac{c_2 \cdot (1 - 2y_1) - c_1 \cdot (1 - 2y_2)}{y_2 - y_1} = \left(\frac{c_2 - c_1}{y_2 - y_1} \right)^2 + 2 \cdot \frac{c_2 - c_1}{y_2 - y_1} \end{aligned}$$

$$(\pi_H - \pi_L)(y_1 - y_2) \geq c_1^2 - y_2 \cdot \left(\left(\frac{c_2 - c_1}{y_2 - y_1} \right)^2 + 2 \cdot \frac{c_2 - c_1}{y_2 - y_1} \right) - \left(\frac{c_1 \cdot y_2 - c_2 \cdot y_1}{y_2 - y_1} \right)^2$$

I could try to make this prettier and would like to know a way to do so (but I've used a lot of time on it and honestly don't want to).

If this equality holds the principal prefers the first best solution getting a_1 from the Agent

- e) : since it is always more expensive to enforce a more "expensive" action in second best then if it is still preferred to do in second best it must also be preferred in first best